

Assignment: Linear Regression (Hands-on Exercise)

Please create a **synthetic dataset** and implement a **linear regression model end-to-end**. The objective is to clearly understand each step and work with realistic data challenges.

Dataset Requirements

Your dataset should include:

- Numerical feature(s) suitable for linear regression
- At least one categorical variable
- Missing values (introduced intentionally)
- A few outliers

Steps to Follow

1. Data Creation

- Generate a synthetic dataset based on a real-world scenario.

2. Exploratory Data Analysis (EDA)

- Visualize the data.
- Identify correlations, missing values, outliers, and relationships between variables.
- **Note:** Understand what to do when variables are highly correlated and implement the necessary changes.

3. Data Preprocessing

- Handle missing values.
- Encode categorical variables.
- Treat or justify the handling of outliers.
- Split the dataset into training and testing sets.

4. Model Building

- Implement a linear regression model step by step.

5. Model Evaluation

- Evaluate the model using appropriate metrics.
- Interpret the coefficients and results.

6. Key Observations

- Summarize learnings from each step of the process.

Expectation

Focus on **understanding and explanation** at every stage. Clarity of reasoning is more important than complexity.